

# Submission Report: Hermes by TheKade

## Intelligent Document Assistant Ashen Era Archive

**Track:** Sub-track 1C (with full foundational support for 1A and 1B)

**Team:** TheKade

**Date:** 09th September 2026

## 1. Problem Statement and Chosen Sub-Track

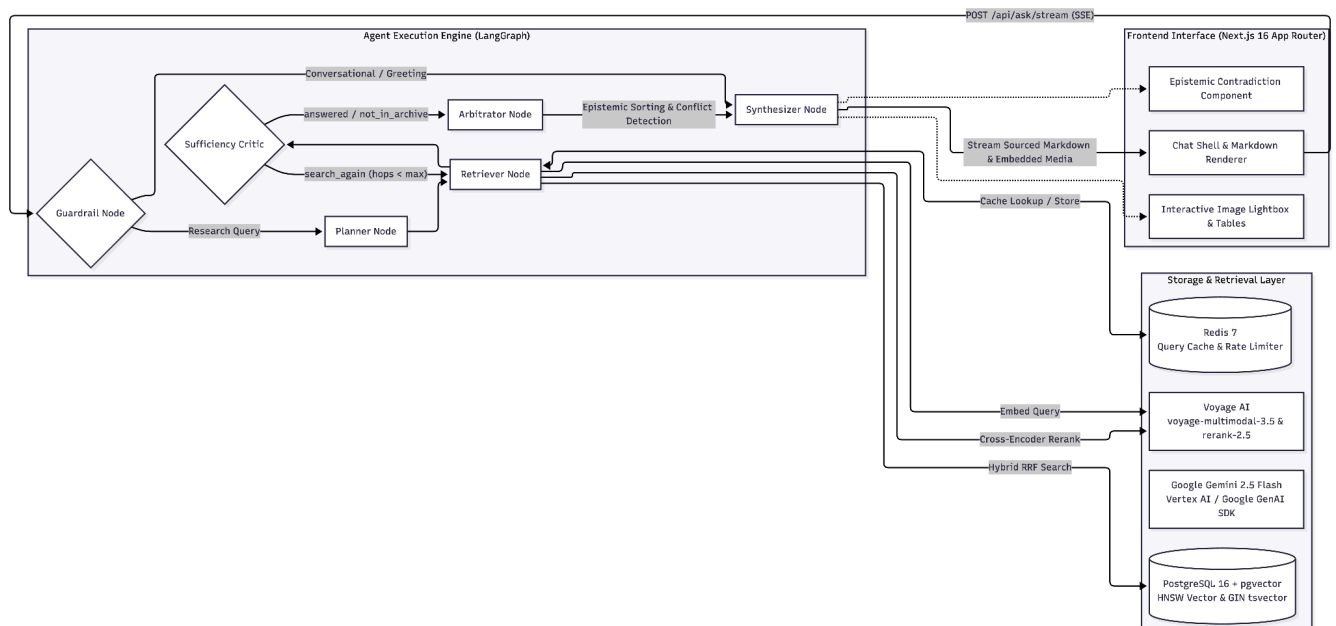
### 1.1 Context and Challenge

Enterprise documentation is fragmented, heterogeneous, and contradictory. The Ashen Era Archive spans 415 documents and 1,277 pages across five formats: codex tables, chronicles, wikis, simulated scans, and figure plates. LLMs lack prior knowledge of this world. A documentation assistant cannot rely on memorized weights; it must discover, connect, arbitrate, and present grounded evidence.

Standard Retrieval-Augmented Generation (RAG) single-lookups fail on enterprise documentation because:

- Multi-Hop Dependencies:** Questions like *"Which war was won by the organization that included Isolde Mournvale?"* or *"In which year was the Gauntlet of Sorrowfell actually forged?"* cannot be resolved in one step. The pivot entity remains unknown until initial document retrieval.
- Synergy with Tracks 1A & 1B:** Solving Sub-track 1C inherently unifies the other two tracks: **Track 1A (Rich Answers):** The iterative loop retrieves visual figure plates and tables, embedding them inline alongside text. **Track 1B (Connecting Facts Across Pages):** Cross-document chains are traversed iteratively across novels, codexes, and ephemera.

## 2. Solution Overview and Architecture



## 2.2 Deliberative Node Responsibilities

| Node               | Function                     | Operational Mechanics                                                                                                                                                                                                                                           |
|--------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Guardrail</b>   | Intent Classification        | High-speed regex classifier. Bypasses retrieval for conversational greetings, routing directly to synthesizer to save quota.                                                                                                                                    |
| <b>Planner</b>     | Contextual Query Planning    | Analyzes question structure, resolves pronouns via dialogue history, identifies primary pivot entities, and formulates 1-2 targeted search terms.                                                                                                               |
| <b>Retriever</b>   | Hybrid Retrieval & Reranking | Embeds queries via <code>voyage-multimodal-3.5</code> , runs SQL Reciprocal Rank Fusion (RRF) across dense HNSW vector and full-text BM25 indexes, and applies <code>rerank-2.5</code> cross-encoder reranking. Results are cached in Redis.                    |
| <b>Critic</b>      | Sufficiency Evaluation       | Inspects accumulated evidence against the goal. Returns <code>answered</code> (sufficient evidence), <code>not_in_archive</code> (detail missing; prevents budget exhaustion), or <code>search_again</code> (emits <code>next_queries</code> for the next hop). |
| <b>Arbitrator</b>  | Epistemic Conflict Detection | Consolidates multi-hop evidence, sorts passages by epistemic authority tiers, and isolates genuine factual contradictions.                                                                                                                                      |
| <b>Synthesizer</b> | Sourced Streaming Synthesis  | Streams citation-backed Markdown over SSE, embedding visual figures ( <code>![Caption](/assets/file.png)</code> ) and tables inline.                                                                                                                            |

## 3. Key Technical Decisions and Rationale

### 3.1 Relational & Vector Unification via PostgreSQL 16 + pgvector

- **Decision:** Consolidated document metadata, chunk texts, 1024-dimensional dense vectors, full-text inverted indexes, session history, and LangGraph persistent state checkpoints (`AsyncPostgresSaver`) into PostgreSQL 16 with `pgvector`, avoiding standalone vector stores (Pinecone, Qdrant, Milvus).
- **Rationale:** Following KISS and DRY principles, a single relational engine eliminates multi-database synchronization overhead, dual-writes, and vector SaaS vendor lock-in.

### 3.4 Ingestion-Time Multimodal Scene Indexing

- **Decision:** Pre-processed figure plates and scanned documents with Gemini Vision API during offline ingestion, generating structured markdown evidence blocks (`title`,

`extracted_text`, `visual_description`, `attributes`) cached in `visual_catalog.json` and chunk payloads.

- **Rationale:** Dynamic runtime image querying introduces 10-15s latency per plate. Pre-extracting verbatim labels, gauge readings, and visual scenes into chunk metadata enables standard hybrid vector search to find visual plates directly via text queries (e.g., *"serpent motif on Gauntlet of Sorrowfell"*).

### 3.5 Epistemic Authority Hierarchy & Discrepancy Assumptions

- **Decision:** Implemented a four-tier authority weighting scheme
  1. **Codex & Figure Plates (1.0 - Canon):** Official gazetteers, anatomical plates, and technical data tables represent author-defined ground truth.
  2. **Wiki Articles (0.8 - Curated Consensus):** Encyclopedic secondary summaries compiled by historians; accurate on broad facts, but prone to editorial drift.
  3. **Novels & Chronicles (0.6 - Narrative Subjectivity):** First- and third-person accounts shaped by character biases, dramatic tension, and unreliable narrators.
  4. **Ephemera (0.4 - Fragmentary Evidence):** Tavern ballads, confiscated ledgers, personal letters, and hurried compacts containing localized rumors and hearsay.
  5. **Contradiction Transparency:** When records conflict (e.g., founding years or troop counts), naive RAG hallucinates or averages numbers. Explicit weights allow stating canonical facts while acknowledging secondary conflicting accounts in contradiction banners.

### 3.6 Fail-Open Architecture and Checkpoint Resilience

- **Decision:** Configured Redis and PostgreSQL checkpointers to fail open. If Redis fails, caching and rate limiting are bypassed without errors. If PostgreSQL checkpointer setup fails, LangGraph falls back to an in-memory `MemorySaver`.
- **Rationale:** A documentation assistant must remain usable during partial degradation. Transient cache outages must not cause user-facing 500 errors.

## 4. What Works, What Doesn't, and Limitations

### 4.1 What Works Reliably

1. **Multi-Hop Traversal:** Reliably resolves multi-hop questions (e.g., identifying a character's faction in Hop 1, then retrieving its war victory in Hop 2).
2. **Rich Embedded Media (Sub-track 1A):** Answers embed visual plates, diagrams, and markdown tables inline with interactive lightbox modals and citation pills.
3. **Contradiction Arbitration:** Isolates true factual contradictions (e.g., founding year of Gloamreach: Codex 246 AS vs. Wiki 67 AS vs. Ephemera 286 AS) and surfaces them transparently without picking a silent winner.

## 4.2 Approaches Tried That Failed

| Failed Approach                  | Observation / Failure Mode                                                                                       | Replacement Solution                                                                                                                          |
|----------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pure Dense Vector Search</b>  | Missed exact proper nouns, item codes, and numeric ratings (e.g., attunement costs or garrison numbers).         | Replaced with SQL Reciprocal Rank Fusion (HNSW vector + tsvector BM25) followed by cross-encoder reranking.                                   |
| <b>Unconstrained ReAct Agent</b> | Model frequently drifted off-topic, repeated identical search queries with synonyms, and exceeded token budgets. | Replaced with a stateful LangGraph pipeline featuring an explicit Sufficiency Critic with a hard budget cap ( <code>MAX_SEARCH_HOPS</code> ). |

## 4.3 Realistic Limitations & Operational Assumptions

1. **Offline Repository-Bound Ingestion (No Ad-Hoc Web Upload)**: Adding documents requires placing them in `data/raw_archive` and running the offline CLI worker (`make ingest`). Heavy document conversion (Docling), multimodal analysis (Gemini Vision), SHA-256 deduplication, and HNSW indexing require compute that cannot run safely within request lifecycles without risking server starvation.
2. **External Provider Quotas & Rate Limiting**: Concurrent batch testing against benchmark questions risks triggering external API rate limits (HTTP 429 on Vertex AI or Voyage AI). Redis caching and exponential backoff prevent repeated query thrashing, but evaluations remain bound to provider RPM/TPM ceilings.

# 5. AI Usage & Governance Disclosure

## 5.1 AI Tools & Models Utilized

| Tool / Service                 | Model / Version                        | Operational Scope                                                                                                                                                 |
|--------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Google Antigravity</b>      | Agentic Coding Assistant               | End-to-end repository development, architectural implementation, LangGraph workflow wiring, Next.js frontend UI components, refactoring, and AST static analysis. |
| <b>Google Gemini</b>           | Gemini 3.8 Flash High / Flash Thinking | Domain analysis over the Ashen Era Archive, RAG retrieval strategy formulation, prompt experimentation, and epistemic weighting calibration.                      |
| <b>Google Gemini (Runtime)</b> | <code>gemini-2.5-flash</code>          | Live execution of LangGraph nodes (Planner, Critic, Arbitrator, Synthesizer) and offline multimodal document/plate extraction ( <code>vision.py</code> ).         |

| Tool / Service | Model / Version                    | Operational Scope                                                                          |
|----------------|------------------------------------|--------------------------------------------------------------------------------------------|
| Voyage AI      | voyage-multimodal-3.5 & rerank-2.5 | Dense 1024-dimensional multimodal vector embeddings and cross-encoder candidate reranking. |
| OpenRouter API | google/gemini-2.0-flash-exp:free   | Fallback model provider configured for non-blocking asynchronous execution.                |

## 5.2 AGENTS.md Compliance & Quality Assurance

All AI-assisted development strictly adhered to **AGENTS.md** engineering conventions:

- **KISS, DRY, SOLID:** Single-responsibility graph nodes; no speculative future-proofing.
- **Zero Dead Code:** AST scans confirmed 0 unused functions, variables, or imports across all backend modules.
- **Read-Only Data Integrity:** The raw archive (`data/raw_archive/`) was treated as strictly read-only throughout ingestion and testing.
- **Auditability:** Complete conversational trajectories and development transcripts are preserved in `ai_usage/` (`chat_*.txt`), reproducible on demand via `make export-chat`.

## Team Contributions

- **Nayantha Nethsara:** Developed the Agent (mostly backend)
- **Shalon Fernando:** Did initial research and RAG system foundation
- **Helindu Senadeera:** Frontend Development
- **Manula Cooray:** Researched the solution and bug fixes